



GeoSosEkon: An Multimodal Integrated Spatial-Temporal Analytics System for Modeling Provincial Poverty Dynamics in Indonesia Using UMAP, PCA, GMM, Ensemble Forecasting, SHAP, Moran's I Analysis, LISA, Fixed Effect Panel Data Regression and Sentiment Analysis using X with RoBERTa

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Abstract. Poverty remains one of the most persistent structural challenges in Indonesia, with significant disparities across its 38 provinces driven by heterogeneous socioeconomic conditions. This project presents GeoSosEkon, a multimodal integrated spatial-temporal analytics system designed to model provincial poverty dynamics in Indonesia using data spanning 2021 to 2024 sourced from the Badan Pusat Statistik (BPS). The system employs a fully parallel modular pipeline comprising six independent components: dimensionality reduction and clustering via both UMAP and PCA compared against Gaussian Mixture Models (GMM) to identify probabilistic provincial socioeconomic typologies; ensemble time-series forecasting to project poverty trajectories at the provincial level; adaptive SHAP (SHapley Additive Explanations) for model interpretability and feature attribution; spatial autocorrelation analysis through Moran's I and Local Indicators of Spatial Association (LISA) to detect geographic clustering and spillover effects of poverty; Two-Way Fixed Effect Panel Data Regression to establish causal inference between socioeconomic indicators and poverty rates while controlling for province-level and time-level heterogeneity; and RoBERTa-based sentiment analysis on 521 poverty-related posts scraped from X (formerly Twitter) spanning 2021 to 2026 as a complementary qualitative insight layer. All analytical outputs are integrated into an interactive web dashboard for policy-oriented visualization and exploration. Findings from this system are expected to provide actionable, data-driven insights for targeted regional poverty alleviation strategies in Indonesia.

Keywords: provincial poverty, spatial autocorrelation, UMAP, Gaussian Mixture Model, fixed effect panel regression, SHAP, LISA, Moran's I, ensemble forecasting, sentiment analysis, RoBERTa, Indonesia, socioeconomic analytics

CHAPTER 1 INTRODUCTION

Poverty remains a deeply entrenched structural challenge in Indonesia, a nation of over 270 million people spread across 38 provinces with pronounced socioeconomic heterogeneity. Recent data indicate that provincial poverty rates range from 4.53% to 26.80%, while human capital indicators such as ICT skills vary from 29.82% to 92.36%,



reflecting long-standing structural disparities in the Indonesian developmental trajectory [1]. Despite consistent poverty reduction trends observed over the past decade, significant inequality persists among provinces, particularly in the eastern regions of Indonesia [2]. These disparities are not merely economic in nature, they are deeply spatial. Spatial autocorrelation analysis has shown that the distribution pattern of poverty across Indonesian regencies and cities is clustered rather than random, forming what researchers term poverty pockets that do not correspond to government administrative boundaries, thus necessitating integrative rather than uniform regional policy approaches [3]. This geographic concentration implies that poverty alleviation strategies must account for cross-provincial interdependencies and spillover effects that standard non-spatial analytical frameworks cannot capture.

To address these analytical gaps, machine learning methods have been increasingly adopted for poverty modeling. Studies applying machine learning techniques to forecast poverty at the district and city level using BPS socioeconomic data have demonstrated the viability of data-driven approaches to poverty prediction in Indonesia [4]. Among ensemble methods, Random Forest and Extreme Gradient Boosting (XGBoost) have been applied specifically to forecast poverty as supervised ensemble learning classification approaches, owing to their robustness and interpretability in socioeconomic contexts [5]. The choice of dimensionality reduction technique prior to clustering also critically affects analytical outcomes. Comparative studies across 240 synthetic datasets have shown that UMAP outperforms PCA in clustering quality, with UMAP achieving a mean Silhouette coefficient of 0.73 compared to 0.46 for PCA, while PCA shows a negative association between performance and the number of clusters to recover [6]. Nevertheless, PCA retains advantages in interpretability and stability for small sample sizes, motivating the need for empirical comparison rather than a priori selection. Alongside these clustering approaches, causal inference remains essential. Two-way fixed-effects panel modeling is widely regarded as the gold standard for leveraging longitudinal data structure to infer causality from within-unit effects, effectively controlling for time-invariant heterogeneity across observational units [7].

Beyond quantitative tabular analysis, public discourse on poverty increasingly manifests on social media platforms, offering a complementary layer of qualitative insight into societal perceptions of poverty dynamics. RoBERTa-based models have demonstrated strong performance for multi-class sentiment and emotion classification on Indonesian Twitter data, establishing IndoRoBERTa as an effective architecture for Indonesian social media text analysis [8]. Integrating such sentiment signals alongside spatial, econometric, and ensemble machine learning components represents the core methodological contribution of this project. GeoSosEkon therefore synthesizes UMAP and PCA comparison for GMM clustering, ensemble forecasting with Random Forest, XGBoost, and LightGBM, adaptive SHAP for interpretability, Moran's I and LISA for spatial autocorrelation, two-way fixed-effect panel regression for causal inference, and RoBERTa-based sentiment analysis on poverty-related posts scraped from X (formerly Twitter) into a unified multimodal spatial-temporal analytics system for modeling provincial poverty dynamics in Indonesia from 2021 to 2024.

CHAPTER 2 RESEARCH METHODOLOGY

A. Dataset Explanation

This study utilizes two distinct datasets reflecting the multimodal nature of the analytical framework. The primary dataset is tabular socioeconomic data sourced from Badan Pusat Statistik (BPS), comprising 151 observations across 38 provinces of Indonesia from 2021 to 2024, with 2021 containing 37 observations due to one missing provincial record. The dataset encompasses 16 variables across four thematic groups: education (School Participation Rate for age brackets 13–15, 16–18, and 19–24 years), labor market (Open Unemployment Rate and Labor Force Participation Rate, each measured in February and August), poverty (Poverty Line in Rupiah, Number of Poor Population in thousands, and Percentage of Poor Population, all measured in March and September, with poverty percentage ranging from 4.00% to 32.97%), and human development (Human Development Index ranging from 61.4 to 83.55, Mean Years of Schooling, and Expected Years of Schooling). The dependent variable throughout this study is the Percentage of Poor Population, with the dual March–September measurement structure addressed in the preprocessing stage of each analytical notebook.

The secondary dataset consists of 521 tweets scraped from X (formerly Twitter) using the keyword "persentase penduduk miskin," covering 2021 to 2026 with a distribution of 94, 100, 100, 87, 100, and 40 tweets per year respectively. All tweets are in Indonesian and contain text, timestamp, engagement metrics, and user metadata across 19 columns. Since the location field contains no non-null values across all records, province-level geospatial attribution of sentiment is not feasible. This dataset is therefore used exclusively as a national-level temporal sentiment layer, with RoBERTa-based analysis producing monthly and yearly sentiment trends that serve as a qualitative complement to the spatial and econometric findings from the BPS data.

B. Research Variable

Table 1. Research Variables

Variable Name	Explanation
APS 13-15	Angka Partisipasi Sekolah 13–15 (School participation rate for ages 13–15, showing the proportion of adolescents enrolled in education)
APS 16-18	Angka Partisipasi Sekolah 16–18 (School participation rate for ages 16–18, indicating the percentage of older teens attending school)
APS 19-24	Angka Partisipasi Sekolah 19–24 (School participation rate for ages 19–24, reflecting youth participation in higher education or vocational training)
TPT - Februari	Tingkat Pengangguran Terbuka Februari (Open unemployment

	rate in February, representing the share of the labor force without work but actively seeking employment)
TPT - Agustus	Tingkat Pengangguran Terbuka Agustus (Open unemployment rate in August, capturing seasonal or mid-year unemployment trends)
TPAK - Februari	Tingkat Partisipasi Angkatan Kerja Februari (Labor force participation rate in February, showing the proportion of the working-age population engaged or seeking work)
TPAK - Agustus	Tingkat Partisipasi Angkatan Kerja Agustus (Labor force participation rate in August, reflecting workforce activity during mid-year)
Persentase Penduduk Miskin - Maret	Poverty rate in March, indicating the percentage of people living below the poverty line
Persentase Penduduk Miskin - September	Poverty rate in September, showing seasonal variation or mid-year poverty levels
Indeks Pembangunan Manusia	Human Development Index, summarizing achievements in health, education, and income
Harapan Lama Sekolah (Tahun)	Expected years of schooling, estimating the total years a child entering school is expected to spend in education
Rata-Rata Lama Sekolah Penduduk Umur 15 Tahun ke Atas Menurut Provinsi	Average years of schooling for adults aged 15+, reflecting educational attainment of the adult population
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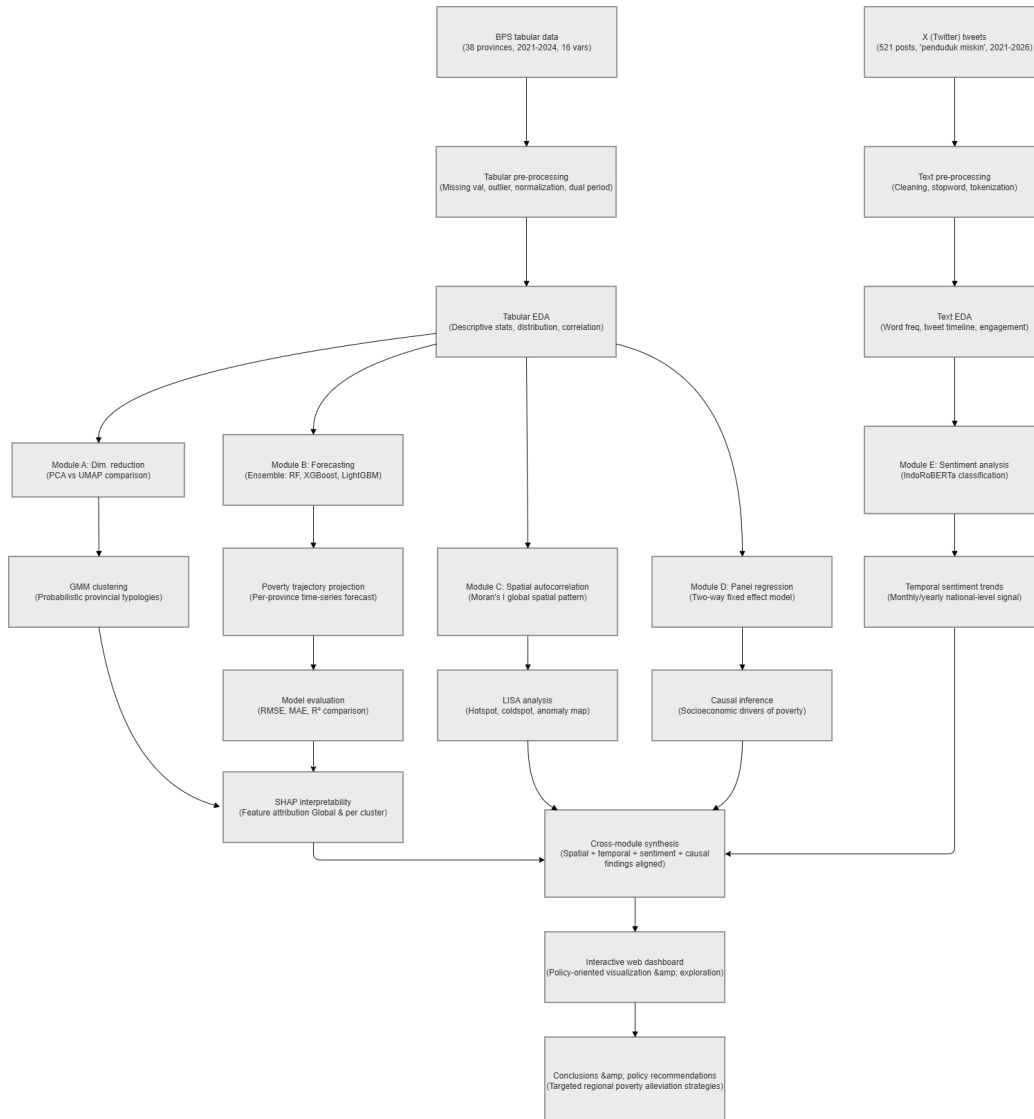
C. Research Workflow

The pipeline begins with two parallel ingestion streams: tabular socioeconomic data sourced from BPS and text data scraped from X (formerly Twitter). Each stream undergoes independent preprocessing and exploratory data analysis before entering the analytical modules. The BPS stream is preprocessed through missing value imputation, outlier handling, normalization, and resolution of the dual March–September measurement structure, followed by descriptive statistical analysis and correlation-based EDA. The Twitter stream undergoes text cleaning, stopword removal, and tokenization, followed by word frequency analysis and tweet timeline visualization.

Following preprocessing and EDA, five independent modules operate in parallel on the BPS tabular data dimensionality reduction and clustering (Module A), ensemble forecasting (Module B), spatial autocorrelation analysis (Module C), two-way fixed effect

panel regression (Module D), and SHAP-based interpretability (integrated across Modules A and D) while Module E conducts RoBERTa-based sentiment analysis on the Twitter stream concurrently. SHAP is applied as a unified interpretability layer, producing global feature attribution from the regression model and per-cluster attribution from the GMM clustering output. All module outputs are subsequently integrated in a cross-module synthesis stage and rendered into an interactive web dashboard for policy-oriented exploration.

Figure 1. GeoSosEkon Research Flowchart



D. PCA-UMAP And GMM

Principal Component Analysis (PCA) and Uniform Manifold Approximation and Projection (UMAP) are used as dimensionality reduction techniques before clustering. PCA transforms the original variables into orthogonal principal components maximizing variance, represented as $z = Xw$ where w is the eigenvector of the covariance matrix. UMAP preserves local and global manifold structure by optimizing cross-entropy between

high-dimensional and low-dimensional representations. Both methods reduce the 16 socioeconomic indicators into a 2–3 dimensional embedding space, then evaluated using Silhouette Score, Davies-Bouldin Index, and BIC to identify the most separable cluster structure before applying Gaussian Mixture Models (GMM) [9]. GMM models the data as a probabilistic mixture of Gaussian distributions with likelihood function $L(\mu, \Sigma, \pi) = \prod_{n=1 \rightarrow N} \sum_{k=1 \rightarrow K} \pi_k \cdot N(x_n | \mu_k, \Sigma_k)$ where π_k , μ_k , and Σ_k denote the mixing weight, mean, and covariance of the k -th cluster. Parameters are estimated using the Expectation-Maximization algorithm, enabling soft cluster assignment and capturing socioeconomic ambiguity between provinces.

E. Ensemble Forecasting + SHAP

Ensemble forecasting is implemented using Random Forest, XGBoost, and LightGBM to predict provincial poverty rates from panel socioeconomic data. Predictions from each model are combined using weighted averaging based on cross-validation RMSE performance. Random Forest constructs multiple bootstrap decision trees, while XGBoost and LightGBM iteratively minimize residual errors through gradient boosting, with LightGBM applying histogram-based leaf-wise growth for computational efficiency. Forecasting is conducted for each provincial cluster to generate poverty projections for 2025–2026 [10]. Model interpretation is performed using SHAP (SHapley Additive Explanations), where each feature contribution is calculated as $\phi_k(f, x) = \sum_{S \subseteq F, k \in S} \frac{(|S|-1)! \cdot (|F|-|S|)!}{|F|!} \cdot [f(S \cup \{k\}) - f(S)]$, ensuring consistency and fairness in feature attribution. Results are visualized through beeswarm and waterfall plots to identify the most influential socioeconomic indicators for each province [11].

F. Spatial Autocorrelation

Spatial autocorrelation analysis is conducted using Global Moran's I and Local Indicators of Spatial Association (LISA) to examine geographic clustering of poverty across Indonesian provinces. Global Moran's I is defined as $I = \frac{n}{\sum_i \sum_j w_{ij}} \cdot \frac{(\sum_i \sum_j w_{ij} (x_i - \bar{x})(x_j - \bar{x}))}{(\sum_i (x_i - \bar{x})^2)}$ where w_{ij} represents the spatial weight matrix based on queen contiguity. Positive values indicate clustered patterns, values near zero indicate randomness, and negative values indicate dispersion, with significance tested using 999 permutations [12]. LISA decomposes the global statistic into local province-level contributions using $I_i = ((x_i - \bar{x}) / m_i) \cdot \sum_j w_{ij} (x_j - \bar{x})$ allowing provinces to be classified into High-High, Low-Low, High-Low, and Low-High spatial regimes for identifying poverty concentration and regional anomalies.

G. Fixed Effect Model Panel Regression

Two-Way Fixed Effect (TWFE) panel regression is employed to estimate the relationship between socioeconomic indicators and provincial poverty rates while controlling for unobserved heterogeneity across provinces and years. The regression model is specified as $y_{it} = \alpha_i + \lambda_t + \beta X_{it} + \varepsilon_{it}$ where α_i captures province fixed effects, λ_t captures year fixed effects, and X_{it} represents time-varying socioeconomic variables [13]. Estimation uses within-transformation by demeaning variables across province and year dimensions.

Standard errors are clustered at the province level to address serial correlation, while model selection between Fixed Effect and Random Effect is validated using the Hausman Test. Final outputs include coefficient estimates, clustered standard errors, t-statistics, p-values, and within-R².

H. Sentiment Analysis RoBERTa

Sentiment analysis is conducted on 521 poverty-related tweets using IndoRoBERTa, a RoBERTa-based pretrained language model optimized for Indonesian text. RoBERTa improves BERT through dynamic masking, larger training batches, and removal of the Next Sentence Prediction objective. Tweet text is tokenized into subword units and processed through a classification head producing sentiment probabilities via the softmax function $P(c | x) = e^{z_c} / \sum_c e^{z_c}$ where z_c is the logit score for class c . IndoRoBERTa achieved 98% accuracy and 97.4% F1-score on Indonesian Twitter sentiment classification tasks [14]. The pipeline assigns sentiment labels and confidence scores to each tweet, then aggregates the results by month and year to produce a temporal trend visualization of public discourse on poverty from 2021–2026 as a complementary qualitative analysis layer.

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Appendix

[Drive Documentation](#)

Member Contribution

Name	Contribution
Rizki Piji Fathoni	Flowchart , Chapter 2C (Research Workflow) , Document structure & consistency



Muhammad Raffi Fahrezi	PPT, Chapter 1 (Introduction) , References (APA formatting)
Nazril Ravi Pratama	BPS data & preprocessing , Twitter dataset , Chapter 2A & 2B (Dataset & Variables)